

WAFRA, Kuwait

WMO: 405920

Lat: 28.567N

Lon: 48.067E

Elev: 351

StdP: 14.51

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.9	42.6	14.7	11.7	56.3	18.3	14.0	60.5	18.8	59.6	17.5	60.9	5.2	280	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	117.9	70.5	116.1	70.3	114.3	70.0	82.9	92.2	80.1	94.7	76.9	94.6	9.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	160.9	89.7	76.2	138.5	88.8	71.9	119.6	86.1	47.4	92.2	44.1	94.7	40.7	94.7	88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	17.0	15.1	DB	34.3	121.6	2.3	1.1	32.7	122.4	31.3	123.1	30.1	123.7	28.4	124.5	(4)
(5)			WB	30.3	85.4	3.4	2.0	27.9	86.9	25.9	88.0	24.0	89.1	21.5	90.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	57.0	61.6	69.9	79.4	91.1	99.1	102.0	101.0	94.5	84.8	69.2	60.2		
	DBStd	16.87	5.79	6.16	6.35	5.70	4.88	3.61	2.82	2.73	3.66	4.73	6.11	5.12		
	HDD50	14	11	2	0	0	0	0	0	0	0	0	0	1		
	HDD65	586	254	125	21	0	0	0	0	0	0	0	26	160		
	CDD50	11303	228	327	615	881	1273	1472	1613	1582	1336	1080	577	319		
	CDD65	6400	6	30	171	431	808	1022	1148	1117	886	615	153	13		
	CDH74	99675	42	242	1569	4758	12293	17830	20559	19512	14049	7491	1235	95		
	CDH80	70677	2	59	619	2437	8228	13521	16095	15049	9871	4363	425	8		
(14) Wind	WSAvg	6.8	6.4	7.0	6.9	6.8	6.9	8.6	8.5	6.9	6.2	5.4	5.6	6.1		
(15) Precipitation	PrecAvg	7.2	1.4	0.9	1.2	0.7	0.2	0.0	0.0	0.0	0.0	0.2	1.3	1.2		
	PrecMax	11.7	4.1	3.2	3.8	2.0	0.8	0.1	0.1	0.1	0.0	1.0	7.2	3.5		
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.4	1.0	0.8	0.9	0.6	0.2	0.0	0.0	0.0	0.0	0.2	1.5	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.2	87.3	100.8	105.0	113.4	119.4	120.6	119.7	114.9	107.7	94.8	81.1		
		MCWB	57.7	59.4	62.7	65.7	68.7	70.1	70.5	71.3	71.3	67.7	62.7	58.2		
	2%	DB	74.2	80.0	91.8	99.6	110.4	117.0	118.2	117.1	111.6	104.0	88.8	76.4		
		MCWB	57.0	57.4	60.4	64.1	68.0	69.7	70.8	71.2	69.8	67.0	62.1	58.8		
	5%	DB	70.7	76.4	86.2	95.9	107.1	114.5	116.2	115.2	109.6	100.8	84.7	73.3		
		MCWB	55.7	56.2	59.5	62.9	67.1	68.6	70.1	71.1	69.0	66.4	61.6	57.7		
	10%	DB	67.5	73.0	81.9	92.0	104.4	112.2	114.2	113.5	107.3	97.7	81.2	70.6		
		MCWB	54.4	55.2	58.7	62.2	66.6	68.0	69.8	70.8	68.5	66.2	61.0	56.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	63.9	69.0	69.7	74.7	77.3	83.5	85.5	84.1	79.8	72.6	66.6		
		MCDB	69.1	72.6	79.2	83.7	92.6	98.1	92.5	94.0	91.7	87.9	77.5	71.7		
	2%	WB	61.6	62.1	66.6	68.6	72.0	72.4	78.8	83.6	81.8	77.0	69.6	64.3		
		MCDB	67.9	71.3	75.6	84.1	93.9	109.0	95.8	93.0	91.2	84.0	77.4	69.6		
	5%	WB	59.0	60.2	64.2	67.2	70.4	70.5	75.0	81.4	78.5	74.8	66.9	61.8		
		MCDB	65.4	69.4	74.5	84.0	95.6	110.2	101.8	94.6	93.4	84.2	76.0	68.5		
	10%	WB	56.6	58.2	62.2	65.7	69.1	69.2	72.1	77.8	74.5	72.7	64.9	59.3		
		MCDB	64.5	68.4	75.2	83.3	95.9	109.1	109.0	99.6	95.2	83.5	74.6	66.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.2	22.0	23.5	23.3	25.1	26.4	26.0	27.1	27.1	25.3	21.1	20.5		
		MCDBR	23.2	25.7	29.3	28.2	28.6	28.8	27.9	28.4	28.7	28.7	25.6	22.8		
		MCWBR	10.6	10.3	10.4	8.6	8.5	9.2	8.8	9.8	10.0	10.5	10.4	10.2		
		MCDBR	19.1	20.4	20.7	22.2	25.6	29.0	28.8	27.4	25.9	23.2	18.9	17.4		
	5% WB	MCWBR	11.0	9.9	9.8	7.6	8.5	10.2	11.4	12.5	11.6	10.5	9.4	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.508	0.559	0.614	0.701	0.704	0.666	0.673	0.626	0.572	0.558	0.543	0.504		
	taud		1.772	1.670	1.559	1.461	1.462	1.496	1.509	1.595	1.683	1.710	1.735	1.792		
	Ebn at Noon		221	222	221	208	208	215	213	222	228	219	208	214		
	Edh at Noon		60	72	86	97	98	94	92	84	75	69	62	57		
(44) All-Sky Solar Radiation	RadAvg		1103	1354	1681	1836	2101	2336	2235	2134	1923	1548	1141	1007		
	RadStd		74	86	86	98	92	91	60	46	64	67	85	89		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.63	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+232	+245

Nomenclature: See separate page